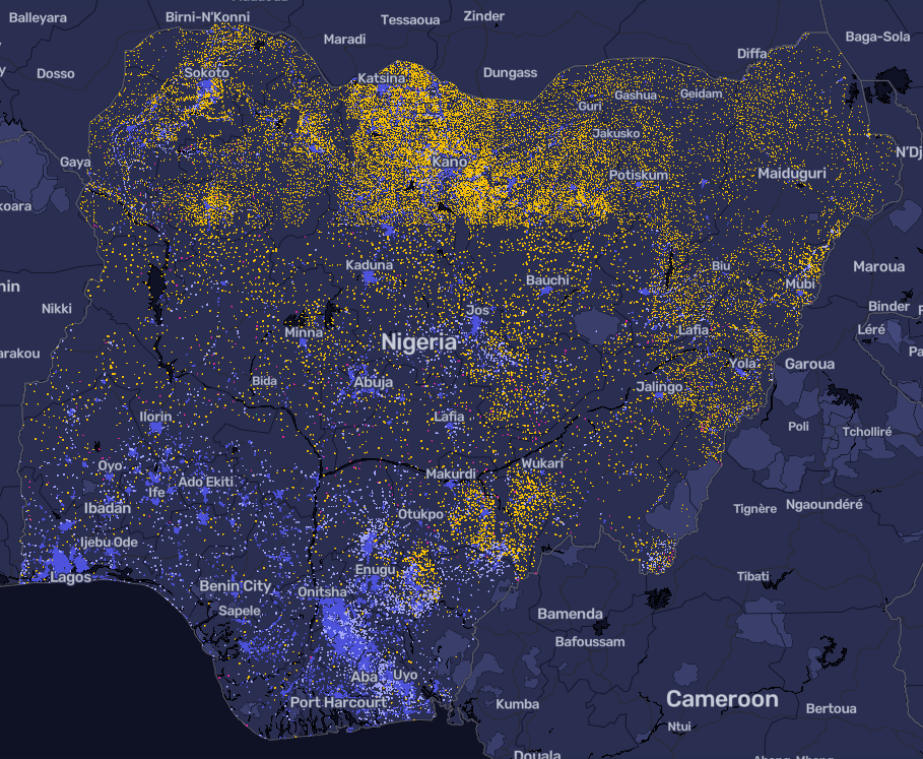


Nigeria

Nigeria OnSSET v2.0

Global Electrification Platform

Explore least cost electrification strategies around the world.



Year

2030

Technologies

- Existing grid
- New grid connection
- Stand Alone PV
- Mini Grid PV Hybrid
- Mini Grid Wind Hybrid
- Mini Grid Hydro

Population connected

Existing grid	163 091 601
New grid connection	40 640 021
Stand Alone PV	57 602 723
Mini Grid PV Hybrid	2 340 812
Mini Grid Wind Hybrid	127
Mini Grid Hydro	33 286
	263 708 570

Investment required

Existing grid	11 176 143 394
New grid connection	7 275 509 517
Stand Alone PV	3 641 376 739
Mini Grid PV Hybrid	201 833 543
Mini Grid Wind Hybrid	26 896
Mini Grid Hydro	3 474 857
	22 298 364 947

Added capacity

Existing grid	4 109 659
New grid connection	1 885 799
Stand Alone PV	498 977
Mini Grid PV Hybrid	52 127
Mini Grid Wind Hybrid	6
Mini Grid Hydro	552
	6 547 120



Scenarios

The model determined the least cost electrification option for each area based on the following assumptions.

Electricity demand target

Bottom-up demand target (Poverty-GDP)

Social & commercial uses

Social and productive uses demand included

Grid generating cost of electricity

Estimated on-grid cost (0.052 \$/kWh)

PV system cost

Expected PV cost

Intermediate investment plan

Capped annual connections

Rollout plan

Nationwide Least Cost approach

Filters

The model results were further narrowed down using the following filters.

No filters applied to the model results.



About the model

Developed by: KTH

Last updated: 2021-04-27

This model is developed using the Open Source Spatial Electrification Tool (OnSSET). Full access to the source code behind this model is available at <https://github.com/global-electrification-platform>. For more information on the functionality of the model please contact the development team at KTH.

